Smart SIEM CTU — Class Diagram

UserRole
READER
ANALYST
ADMIN
SeverityLevel
INFO
WARNING
HIGH
CRITICAL
IncidentStatus
OPEN
IN_PROGRESS
RESOLVED
FALSE_POSITIVE
Correlation Engine
CorrelationScheduler
process(job): void
PassTheHashRule
detect(es, since): DetectionResult[]
BruteForceRule
detect(es, since): DetectionResult[]
LogClearingRule
detect(es, since): DetectionResult[]
DataExfilRule
detect(es, since): DetectionResult[]
CorrelationService
runCycle(): CycleReport
handleDetection(result): void
findRecentDuplicate(result): boolean
pushes alert
runs 4 rules
DetectionRule
detect(es, since): DetectionResult[]
SoarService
NotificationsProcessor
scheduled cycles
SOAR Automation
SoarService
executePlaybook(data): {executionId}
abortPlaybook(executionId): {status}
runPlaybook(name, incidentId): Record
PfSenseClientService
blockIP(ip, reason): {id}
unblockIP(ruleId): void
isolateHost(ip, reason): {ids}
blockPort(ip, port, proto, reason): {id}
listRules(): PfSenseRule[]
checkIP(ip): {blocked, rules}
createAlias(name, addrs, descr): void
deleteAlias(name): void
getSystemInfo(): SystemInfo
request(method, path, body): Response
write(fn): T
HTTP REST
pfSense Firewall
POST /api/v2/firewall/rules
DELETE /api/v2/firewall/rules/{id}
POST /api/v2/firewall/apply
GET /api/v2/firewall/rules
POST /api/v2/aliases
DELETE /api/v2/aliases/{name}
GET /api/v2/system/info
Authentication
JwtAuthGuard
canActivate(context): boolean
RolesGuard
canActivate(context): boolean
ApiKeyGuard
canActivate(context): boolean
AuthService
validateUser(username, password): User | null
signIn(username, password): AuthResponse
signUp(username, password, role?, email?): User
verifyMfa(sessionId, code): AuthResponse
enableMfa(userId, email): void
disableMfa(userId): void
getProfile(userId): User
MfaService
sendOtp(userId): {sessionId: string}
verifyOtp(sessionId, code): userId
enableMfa(userId): void
disableMfa(userId): void
Redis
BullMQ queues: logs
correlation, ueba
notifications, reports
pushes score job
push normalize jobs
Log Ingestion
LogsProcessor
process(job): void
normalizeLogs(rawLogs): NormalizedLog[]
LogsController
ingest(logs): {accepted}
search(query): SearchResult
UebaProcessor
LogsService
ingest(logs: CreateLogDto[]): {accepted: number}
search(query: SearchLogsDto): SearchResult
Dashboard
DashboardController
getCrisisStats(): DashboardStats
getOverview(query): OverviewResult
DashboardService
getStats(): DashboardStats
getOverview(interval, sourceType?): OverviewResult
getSeverityDistribution(): Record
getEventsTimeline(): TimelineBucket[]
getTopSources(): SourceEntry[]
getThreatTypes(): ThreatEntry[]
getLoginFailures(): FailureBucket[]
getSuspiciousActivityHeatmap(): HeatmapCell[]
Reports
ReportsController
generate(dto): {jobId}
status(jobId): JobStatus
download(jobId): Stream
list(): FileList
ReportsProcessor
process(job): void
ReportsService
generate(request): {jobId}
getJobStatus(jobId): JobStatus
download(jobId): {path, filename}
listReports(): FileList
cleanup(): void
fetchLogs(from, to): NormalizedLog[]
ReportGeneratorService
generate(data, request): ReportMeta
generatePdf(data, request): ReportMeta
generateExcel(data, request): ReportMeta
generateCsv(data, request): ReportMeta
getFilePath(filename): string|null
listReports(): FileList
cleanup(): void
Infrastructure
MailService
sendEmail(options): boolean
sendMfaCode(email, code, username): boolean
sendIncidentAlert(incident): boolean
sendDailyDigest(data): boolean
ElasticsearchService
onModuleInit()
bulkInsert(logs: NormalizedLog[]): {indexed: number}
search(query: LogSearchQuery): SearchResult
getClient(): Client
ensureIlmPolicy()
ensureIndexTemplate()
ensureInitialIndex()
PrismaService
user: UserDelegate
incident: IncidentDelegate
correlationRule: CorrelationRuleDelegate
uebaProfile: UebaProfileDelegate
auditTrail: AuditTrailDelegate
mfaSession: MfaSessionDelegate
apiKey: ApiKeyDelegate
send emails
search + index
ORM queries
Google SMTP
smtp.gmail.com:587
MFA OTP delivery
Incident alerts
Daily digest
Elasticsearch
index: ctu-logs-*
search, count, bulk
ILM policy: 30-day retention
PostgreSQL
10 tables via Prisma ORM
Users, Incidents, Rules
UEBA Profiles, Audit Trail
Incidents & Rules
RulesService
findAll(): CorrelationRule[]
findOne(id): CorrelationRule
create(data): CorrelationRule
update(id, data): CorrelationRule
remove(id): void
IncidentsService
findAll(filters): Incident[]
findOne(id): Incident
update(id, data): Incident
Admin & Audit
AuditService
getTrail(filters): AuditEntry[]
verifyBatchIntegrity(batchId): IntegrityResult
AdminService
listUsers(): User[]
createUser(data): User
updateUser(id, data): User
deleteUser(id): void
getRetentionPolicies(): Policy[]
updateRetentionPolicy(data): Policy
Notifications
NotificationsProcessor
process(job): void
handleIncidentAlert(incidentId): void
handleDailyDigest(): void
UEBA
UebaService
findAll(minRisk?, maxRisk?): UebaProfile[]
findOne(userPrincipal): UebaProfile
getStats(): UebaStats
triggerBaselineRebuild(): {usersProcessed}
UebaProcessor
process(job): void
handleScoring(log): void
handleRebuildBaselines(): void
UebaBaselineService
buildAllBaselines(): {usersProcessed}
buildBaselineForUser(user, since): void
UebaScorerService
scoreLog(log): UebaScoreResult
extractFileCount(rawMessage): number
triggerSoar(user, score, log): void
SoarService